

POK MAN (Nathan) HAN

+853 6565 3805 — nathanhan@ucsb.edu — GitHub: hanpokman

Research Interests

Quantum machine learning, superconducting-qubit dynamics, stochastic diffusion models, physics-informed deep learning

Education

University of California, Santa Barbara
Incoming B.S. Computer Science

From Sep. 2026

Pui Ching Middle School, Macau
GPA: 98.17/100

Sep. 2020–Jun. 2026

AP: Microeconomics (5), Statistics (5), Calculus BC (5), Physics 1 (5), Physics C: Mechanics (5), Physics C: E&M (5), Chemistry (5), Computer Science A (4), English Language & Composition (4)

Publication

P. Han, "Streamlining Navigation: Depth-Informed Item Mapping for Enhanced Home Service Robotics," *Proc. 4th Asia-Pacific Conf. Communications Technology and Computer Science (ACCTCS)*, 2024, pp. 587–591. DOI

Research Experience

Research Intern, Prof. Wang Peng's Lab
University of Macau

Jun. 2026–Aug. 2026

- Investigated stochastic diffusion models under the supervision of Prof. Wang Peng.
- Independently developed the Latent Hamiltonian Quantum Propagator (LHQP), a physics-structured sequence model that infers time-varying Pauli-Hamiltonian coefficients from partial observations to forecast coherent drift and crosstalk in small superconducting-qubit systems.
- Evaluated 17 comparison models across 9,150 completed synthetic runs; achieved horizon-mean fidelities of 0.9731 for diagonal drift and 0.9078 for interacting dynamics, with ablations showing that the prescribed interaction terms were necessary for performance.
- Established the method's scope as a state-vector feasibility study and proposed hardware validation using Ramsey and spectroscopy measurements, readout features, and control telemetry, with a future extension for dissipative dynamics.

Aquatic Environment Surveillance Autonomous Surface Vehicle

Jan. 2023–Aug. 2024

- Designed and built an autonomous surface vehicle with a custom 20-meter underwater lowering platform to survey invasive species, water-quality fluctuations, and litter in Macau lakes.
- Served as software lead on a three-person team; owned the edge-AI computer-vision pipeline, engineered the lowering mechanism, and coordinated mechanical integration and the project timeline.

Depth-Informed Item Mapping for Enhanced Home Service Robotics

Sep. 2022–Oct. 2023

- Developed a mapping method using RGB-D data, geometric transformations, and 2D pose estimation for indoor localization and item tracking without a computationally intensive full-SLAM pipeline.
- Served as sole author and lead developer under faculty guidance; formulated the transformations and validated robustness through field testing, culminating in the ACCTCS 2024 publication above.

Technical Skills

Programming: Python, C++, Java; PyTorch, TensorFlow
modeling, computer vision, stochastic modeling

Research: quantum-circuit simulation, GRU/RNN sequence

Languages: English (IELTS 8.5; SAT 1530), Mandarin (native), Cantonese (native)

Selected Awards and Honors

- Special Award, ISEF (2024)
- First Prize, China Adolescents Science and Technology Innovation Contest (2024)
- Macau ISEF Qualification, 8th Place (2026)
- First Prize, Korean Science and Engineering Fair (2024)
- Fourth Prize, Taiwan International Science Fair (2023)
- Asia Pacific ICT Alliance, Merit (2022, 2023)

Leadership and Service

School Robotics Lab (FABLAB), Project Captain

Jun. 2022–Jun. 2026

Led weekly meetings, mentored junior members, and coordinated outreach to engineering professors to guide technical projects.

School Soccer Club, Program Organizer

Sep. 2023–Jun. 2026

Organized recurring intramural matches and expanded participation to more than 20 students.